

BAX 423 — Homework 1 — Part 4

NOAA Storm Events: EF2+ Tornadoes (2020–2025)

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Data source: NCEI Storm Events Database (detail CSV files). Filter: EVENT_TYPE = Tornado, TOR_F_SCALE in EF2–EF5 (plus rare legacy F2–F5). Tie-break for peak month per state: smallest month number if tied. Years: 2020–2025.

Summary counts

Metric	Value
All-event rows loaded (all types, all years)	410,192
EF2+ tornado rows (analysis set)	1,164
Total deaths (DEATHS_DIRECT + DEATHS_INDIRECT)	396
Fatal incident rows (deaths > 0)	137
% of fatal narratives mentioning "mobile home" or "trailer"	50.36% (69 rows)

States with the most EF2+ tornadoes (top 15)

State	Count
MISSISSIPPI	112
TEXAS	111
ALABAMA	93
IOWA	74
TENNESSEE	61
ARKANSAS	61
MISSOURI	60
LOUISIANA	53
ILLINOIS	52
OKLAHOMA	50
KENTUCKY	45
NEBRASKA	40
GEORGIA	39

INDIANA	37
FLORIDA	36

Frequency by month (all states combined)

Month	Count
1	68
2	43
3	254
4	242
5	178
6	81
7	51
8	37
9	12
10	30
11	31
12	137

States where May is the peak month for EF2+ counts

These are the states where, for that state alone, May had the highest count (ties → earliest month). Use this list for the course email (Part 4e).

COLORADO, KANSAS, MICHIGAN, MINNESOTA, TEXAS, VIRGINIA

Regional pattern

Among the highest-frequency states, many are Great Plains, Dixie Alley, or Midwest states where spring and early-summer severe weather is climatologically common. Coastal and western states typically rank lower in raw tornado counts, consistent with the central and southeastern U.S. tornado maxima.

Examples of Plains/Midwest/Southeast among top-10 states: MISSISSIPPI, TEXAS, ALABAMA, IOWA, TENNESSEE, ARKANSAS, MISSOURI, LOUISIANA, ILLINOIS, OKLAHOMA.

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